

A Cyber-Physical-Social Perspective on Future Smart Distribution Systems

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ABSTRACT

An increasing number of distributed energy resources, such as rooftop photovoltaic systems, electric vehicles, and distributed energy storage, are being integrated into distribution systems. The rise of distributed energy resources has come hand-in-hand with large amounts of data generated through data collection, communication, and control devices. In addition, a massive number of consumers are involved in interaction with the power grid to provide flexibility. Electricity consumers, power networks, and communication networks are three main parts of distribution systems and are deeply coupled. In this sense, smart distribution systems can be viewed as cyber-physical-social systems. This article provides a comprehensive review of coupling relationships among cyber, physical, and social aspects of distribution systems. Several emerging topics that challenge future cyber-physical-social distribution systems, including applications of 5G communication, the impact of COVID-19, and data privacy issues, are discussed. The article also envisions future research directions and challenges for cyber-physical-social distribution systems.

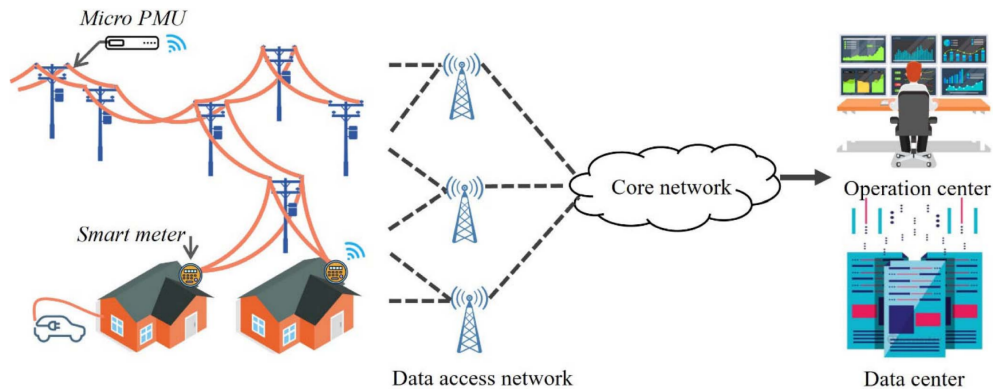


Figure 1: Overview of Cyber-Physical-Social Distribution Systems.